

NOTICE OF CONTRACT ACTION

Program Title: Precise Ensemble Network Communications Allowing PNT (Positioning, Navigation, and Timing) (PENCAP)

Announcement Type: Notice of Contract Action (NOCA)

Solicitation Number: FA2377-26-R-B014

NAICS Code: 541715

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*******This announcement does not request any proposals at this time*******

The following program is an anticipated requirement for the Air Force Research Laboratory, Sensors Directorate, Spectrum Warfare Division (AFRL/RYSW) located at Wright-Patterson Air Force Base, OH. A two-step closed Advanced Research Announcement (ARA) for this requirement is anticipated to be published on SAM.gov around May 2026.

The anticipated funding listed below reflects estimated funding only. This estimate is not a promise of funding. Funding is uncertain and is subject to change. Changes in funding availability may occur at the discretion of the Government.

PROGRAM DESCRIPTION: The purpose of this notice is to provide information about a anticipated two-step closed ARA for Precise Ensemble Network Communications Allowing PNT (PENCAP).

This notice is not a solicitation. The Government will not reimburse costs associated with any communication submitted in response to this notice. No documentation is requested at this time.

TECHNICAL DESCRIPTION: The objective for PENCAP is the development of a network clock ensemble and the implementation of it on a VPX card capable of running pntOS. Prototype cards, networking methods, and an ensemble algorithm will be delivered to the Government for further development and testing.

The algorithm will be designed to be robust in dynamic environments, continuously evaluating both local and remote timing references for stability, trustworthiness, and source validity. It will incorporate a weighting system, allowing references to enter and exit the ensemble without disrupting it, and will feature outlier rejection to discard invalid or unstable measurements, ensuring a consistently accurate time.

To support the algorithm, a resilient network method will be designed to disseminate timing and stability information. This network will utilize various existing communication links on aircraft and can be configured in multiple ways, such as point-to-point or hub-and-spoke. The system will be designed to authenticate new nodes that join the network, ensuring the traceability and integrity of all connections.

The ensemble algorithm and network method will be implemented on a prototype card. This will involve modifying a Commercial-Off-The-Shelf (COTS) 3U VPX card to function with pntOS. The design will emphasize open standards for both data and physical interfaces to simplify future integration and upgrades. The project will culminate in a series of simulated, lab, and flight tests to demonstrate the card's full capabilities.

ANTICIPATED FUNDING: A potential contract is anticipated with an estimated total contract value of \$9,300,000.00. This funding profile is an estimate only and not a contractual obligation for funding. All funding is subject to change due to Government discretion and availability. Potential offerors should be aware that due to unanticipated budget fluctuations, funding in any or all areas may change with little or no notice.

ANTICIPATED CONTRACT TYPE: A Definitive (c-type) Cost-Plus Fixed Fee (CPFF) contract is anticipated.

SET ASIDE: None

**THIS IS NOT A REQUEST FOR PROPOSALS
DO NOT SUMBIT PROPOSALS AT THIS TIME
ANY PROPOSALS RECEIVED WILL BE DISREGARDED**